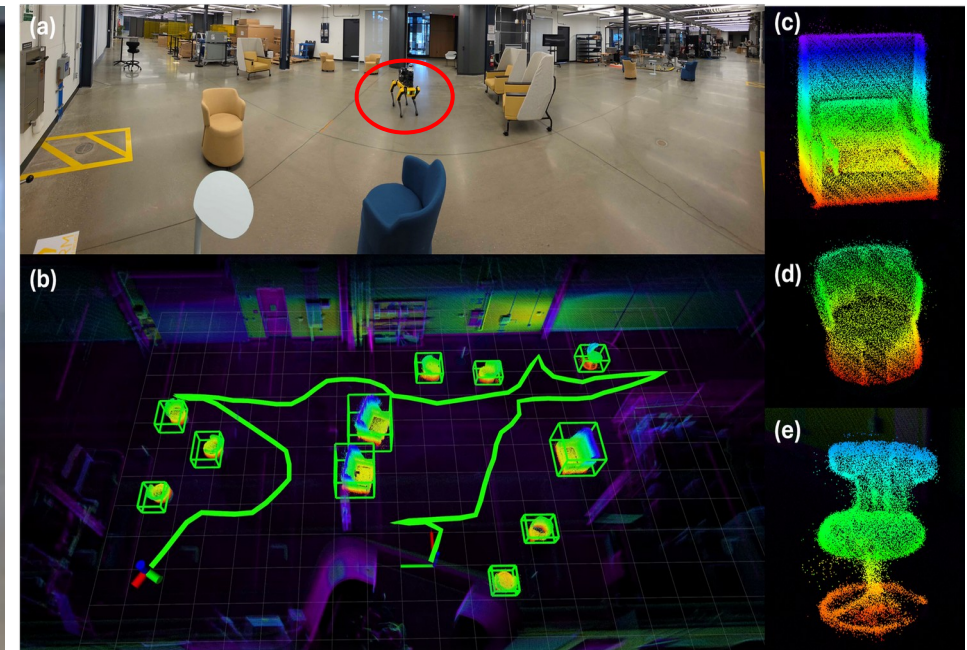


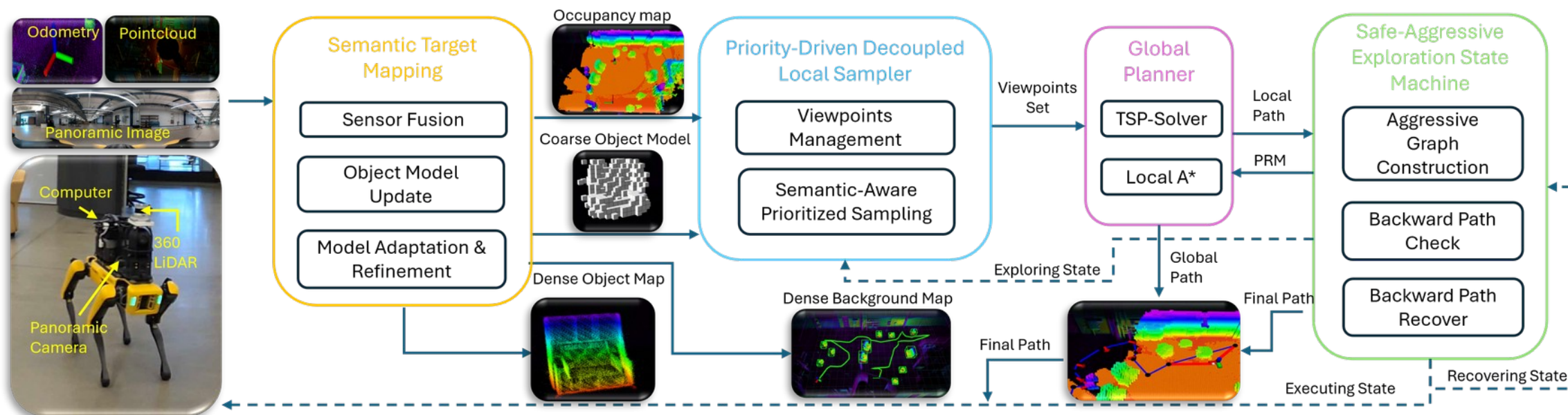
SEDEM: Semantic Exploration and Dense Mapping of Complex Environments using Ground Robot with Panoramic LiDAR-Camera Fusion

Xiaoyang Zhan, Shixin Zhou, Qianqian Yang, Yixuan Zhao, Hao Liu, Srinivas Chowdary Ramineni and Kenji Shimada

**Explore unknown spaces.
Inspect target objects.
Build dense semantic maps.**



Semantic Exploration Planning



Manage semantic target viewpoints and geometric coverage viewpoints separately.
Prioritize object inspection, then cover the remaining space.

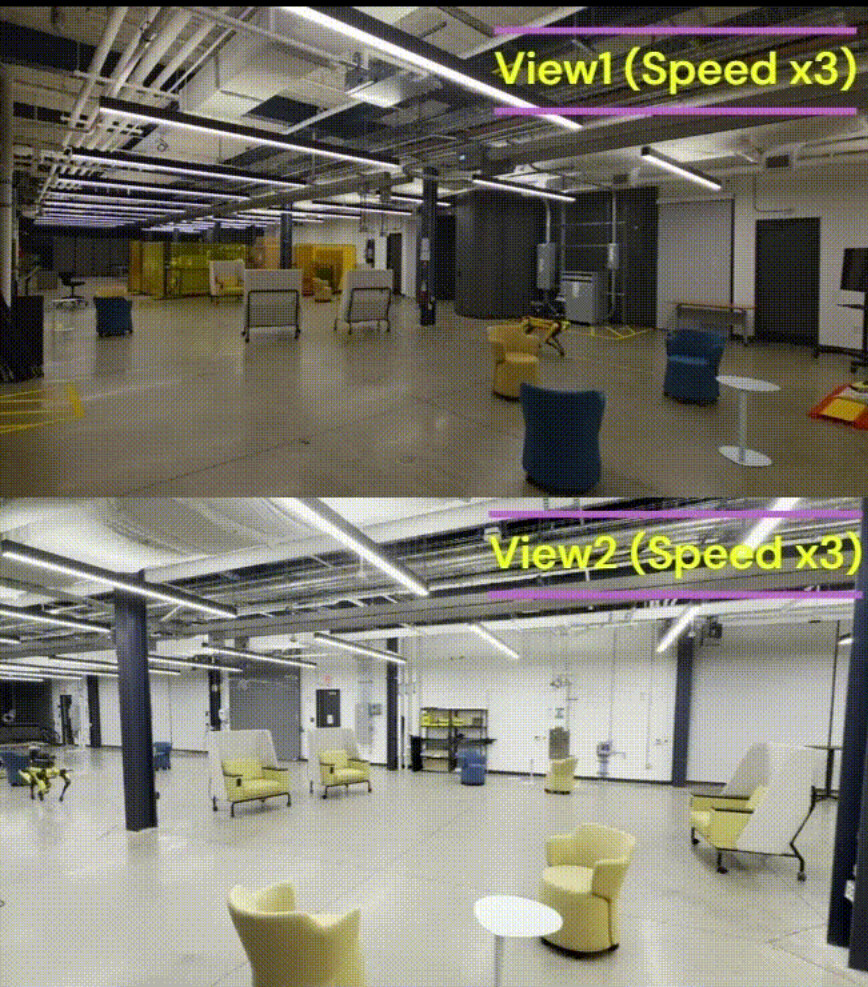
Mock Construction Site

RVIZ View (Speed x3)



5 targets / 54 s exploration

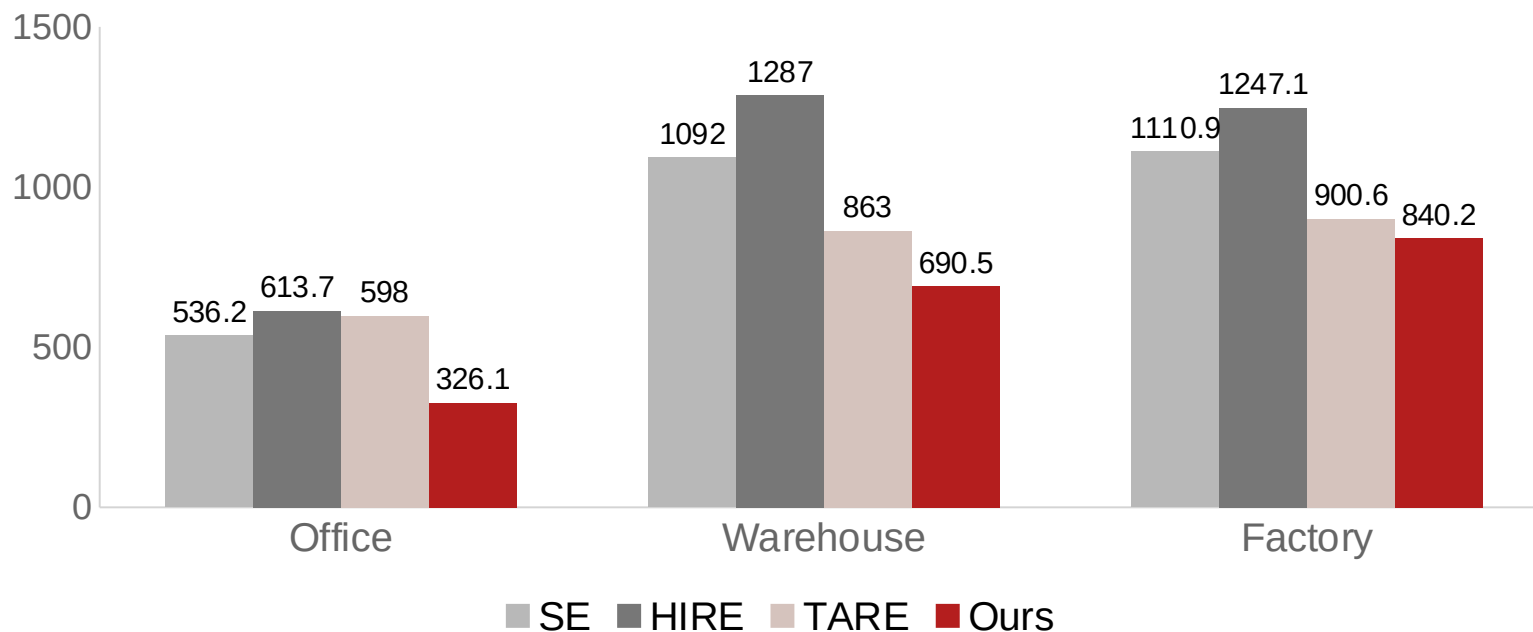
Dense Object Mapping in a Lobby



11 chairs / 1.75 cm mean object-map error

Results and Discussion

Exploration time (s)



Fastest exploration with background mapping and target multi-view observation completed.

Means over 5 runs; error bars omitted. Ground-truth object detections.

